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Original research

Quadriceps torque complexity before and after anterior cruciate ligament reconstruction

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ABSTRACT

Objectives: The purpose of this project was to longitudinally examine quadriceps torque complexity in a group of individuals who tore their ACL and underwent ACL reconstruction.
Design: Cohort analysis.
Methods: Thirty-four individuals completed maximal effort bilateral isometric strength testing after ACL injury but pre-surgery, five months' post-surgery (mid-point of rehabilitation), and when cleared to return to activity. Sample entropy, a nonlinear analysis of quadriceps torque control (complexity), was calculated from maximal isometric contractions. Two 3×2 repeated measures analysis of variance were used to examine changes over time and between limbs for quadriceps torque complexity and peak torque.
Results: Quadriceps peak torque was lower in the involved limb when compared to the uninvolved limb at every time point ($p < 0.001$). Peak torque of the involved limb was decreased at mid-point of rehabilitation compared to before surgery ($p = 0.023$) and at mid-point compared to return to activity ($p = 0.041$). Quadriceps sample entropy was higher in the involved limb compared to the uninvolved limb at the mid-point of rehabilitation ($p < 0.001$) and return to activity ($p < 0.001$), indicating greater complexity. The involved limb also demonstrated increased torque sample entropy from pre-surgery to mid-point of rehabilitation ($p = 0.023$), but not from pre-surgery to return to activity ($p = 0.169$) or from mid-point to return to activity ($p = 0.541$).
Conclusions: Not only does quadriceps strength decline with ACL reconstruction, but quality of the quadriceps muscle contraction is also compromised. Increased torque complexity experienced in the ACL limb after reconstruction may contribute to impaired physical function in individuals following ACL reconstruction.

Practical implications

- Five months after ACL reconstruction surgery, patients have difficulty creating a consistent quadriceps muscle contraction compared to their healthy leg.
- There are negative changes in torque magnitude and complexity at the time in which individuals are cleared to return to sport after ACL reconstruction which could lead to poor movement and might have future injury implications.
- Patients also have decreased strength on their ACL limb compared to their healthy limb after ACL injury and throughout rehabilitation.

1. Introduction

Anterior cruciate ligament (ACL) injuries affect approximately 250,000 young active individuals every year[1,2] and these injuries are commonly treated with surgical reconstruction.[3] Following ACL reconstruction, quadriceps dysfunction (e.g., weakness and atrophy) is regularly present and can persist for years afterwards.[4] As quadriceps dysfunction prevents successful return to physical activity[5] and puts long-term knee joint health at risk,[6] it is of substantial clinical interest.

Quadriceps dysfunction after ACL reconstruction has primarily been quantified by examining the maximal knee extension torque output during isometric contractions.[4] While maximal quadriceps torque (i.e., strength) is an important clinical metric used to identify muscle weakness and is important for creating rehabilitation milestones following ACL reconstruction, it only considers a single point of the torque curve (e.g., the peak). By only assessing the peak torque, information is simply gleaned on the quantity, but not quality, of the torque generated by the muscle. The quality of a muscle contraction is important to understand because it provides information about neuromuscular function[7,8] and muscles' motor unit properties.[9] From a qualitative perspective a high quality muscle contraction would present with a torque curve that shows minimal irregularities and smaller fluctuations of the data points, and would be suggestive of exceptional force control.[10] While a low-quality torque curve would be "noisier" showing many

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fluctuations in torque that are of greater magnitude suggesting less control. Torque quality is also thought to affect overall coordination and motor task performance.[11] For example, Pua et al., found that after ACL injury more irregular quadriceps force control was related to inferior single limb hop performance.[12] Thus, indicating that quadriceps torque quality also likely provides increased clinical knowledge of movement dysfunction. Understanding quadriceps torque quality should allow for further understanding of quadriceps function after ACL reconstruction.

A variety of approaches have been utilized to quantify the quality of a quadriceps contraction. For example, quadriceps force control, measured via root mean square error, has been shown to decrease after ACL injury[13] and 18 months after reconstruction[14] when compared to uninjured control participants. Research has also determined that approximately four years following ACL reconstruction individuals are still presenting with increased quadriceps torque variability, measured via coefficient of variation.[10,15] However, the above-mentioned methods applied to the torque signal to asses quality are limited in that they use linear-based calculations (quantifying torque deviations around a fixed point within a time-series).[16] However, Slifkin and Newell[16] identified that there is an irregular temporal structure in muscle force fluctuations, suggesting that nonlinear analyses are necessary.

Nonlinear analyses assess the deviation of the time series (not around a fixed point) and thus are more sensitive to the fluctuation patterns of the interacting elements (e.g., motoneurons, myofibrils, tendon units) involved in muscle contraction.[17]

Nonlinear analyses have been applied to assess the quality of quadriceps contractions in individuals with ACL injury[12,18] and ACL reconstruction.[19,20] These data show that torque complexity of quadriceps isokinetic maximal contractions (calculated with a wavelet transformation[12] and a frequency analysis in the power domain[18]) is higher in those with an ACL injury compared to their uninvolved limb.[12,18] Similarly, additional work found increased complexity of isometric quadriceps contractions in individuals following ACL reconstruction (measured via approximate entropy), when compared to healthy control participants,[21] Lastly, Czaplicki et al.,[19] found increased quadriceps isokinetic torque complexity, via a wavelet transformation, at six months following ACL reconstruction when compared to healthy control participants. Most of this research has focused on quantifying quadriceps torque complexity during isokinetic strength, though the evaluation of torque quality and its relationship to muscle motor unit properties are best represented when assessing isometric strength.[9] The single paper looking at isometric torque complexity assessed it at a single time point after surgery and thus limits our understanding of how isometric torque complexity changes over time and/or is affected by rehabilitation.[21]

Sample entropy is one form of nonlinear analysis and has not been used previously to characterize torque fluctuations in individuals after ACL reconstruction. Applying sample entropy to the knee extensor torque signal allows one to establish the predictability of the signal between successive data points (i.e., if successive data points are closer in value to each other there would be less complexity/higher quality in the torque signal).[22] This analysis also allows for increased understanding of the torque curve by characterizing the ability of the muscle to have rapid and accurate force adaptation.[7] Thus, using sample entropy, a nonlinear approach to assess torque control, will allow for additional information to be gleaned about the torque time curve compared to previous works that have used linear analyses to assess muscle complexity in isometric torque. From a neurophysiological standpoint, it has been identified that alterations in force fluctuations are regulated by motor unit discharge rates as well as the contractile properties of the motor units,[23,24] in which muscles with few motor units exhibit greater force fluctuations (i.e., increased torque complexity).[23] Following ACL injury and reconstruction there have been documented alterations in muscle fiber conduction velocity, muscle denervation, and changes is muscle fiber typing,[25,26] all of which could contribute to altered muscle quality and increased force fluctuations. Therefore, the purpose of this project was to longitudinally examine quadriceps isometric torque complexity, quantified using sample entropy, in a group of individuals who tore their ACL and underwent ACL reconstruction. We hypothesized that following ACL injury and reconstruction individuals would demonstrate increased torque complexity in the involved limb compared to the uninvolved limb, and that torque complexity would not return to uninjured limb levels by the time of return to activity.

2. Methods

This study is a prospective descriptive cohort study, in which thirty-four individuals who had recently sustained a primary unilateral ACL tear but had not yet undergone reconstructive surgery were recruited to participate. Individuals were enrolled in this study if they were between the ages of 14 and 30 years old, sustained an ACL tear while participating in an organized sport, and were scheduled to undergo ACL reconstruction using an autograft. Individuals were excluded if they had undergone any prior knee surgery, sustained a previous ACL injury, or were expected to receive reconstruction to any other ligament besides the ACL at the time of surgery. All participants and their parent/guardian (if applicable) read and signed an informed consent/assent approved by the University Institutional Review Board (#00110455).

Participants completed three testing sessions, 1) PRE testing 75.6 ± 52.6 days after injury but 5.9 ± 4.0 days prior to surgical reconstruction, 2) Approximately 5.0 ± 0.5 months post-surgery (MID) to assess a midpoint in the rehabilitation process, and 3) at the time that the participant was cleared to return to activity (RTA; 9.4 ± 1.7 months following surgery) by their physician, including a successful completion of a return to sport test. Participants in this study were prescribed the rehabilitation protocol for our clinic, which can be found in Appendix A. The minimal functional RTA criteria in our clinic for patients to be medically cleared to return to physical activity are as follows: 1) Completing an ACL limb single-leg leg press of 15 repetitions with a load of 100 % body weight, 2) A single-leg forward hop for distance recording ≥95 % symmetry ((ACL limb / NonACL limb) ∗ 100) between limbs, 3) A two-legged deep squat for qualitative symmetry, and 4) A single-leg deep squat for qualitative symmetry between limbs in the hip, knee, and ankle, as well as trunk control. These functional criteria are in addition to a physician examination for: 1) appropriate joint laxity, 2) full knee joint range of motion, and 3) no knee joint effusion.

Quadriceps strength testing at all sessions consisted of completing bilateral maximal isometric quadriceps strength tests using an isokinetic dynamometer (Biodex Medical Systems, Shirley, NY, USA). In accordance with manufacturer recommendations, individuals were seated in the dynamometer with their hips and knees flexed to 90°, utilizing cross-body and waist straps. Individuals completed a warmup protocol that consisted of 3 submaximal practice trials with a 2-minute rest between efforts. Participants were then instructed to produce three maximal isometric contractions (MVIC) and hold each of these for 5 s. All participants completed testing on their non-ACL limb first and then subsequently for their ACL limb. During each contraction, verbal encouragement instructing participants to kick out as hard as they can was provided along with real-time visual feedback of their torque curve.

Raw quadriceps torque was recorded with a sampling rate of 2000 Hz from the Biodex via a custom LabVIEW program (LabVIEW version 8.5, National Instruments, Austin TX, USA). Peak quadriceps torque was recorded from the MVIC trial with the largest peak torque value and was normalized to body weight. Using the same trial, sample entropy was calculated as $\mathrm{SampEn}(m, r, \mathrm{N}) = -\ln(A/B)$. Where, m (set to 2 for clinical data)[22] was the number of data points in which the distance between data points was compared to fit within, r, which was the similarity criterion and set to 0.2 times the standard deviation. N was the total window of data, we selected 4000 data points, representing 2 s of the isometric torque curve beginning at the point in which the torque curve plateaus. A represents the number of sequential data points $(m + 1)$ that fall within r, and B is the number of sequential

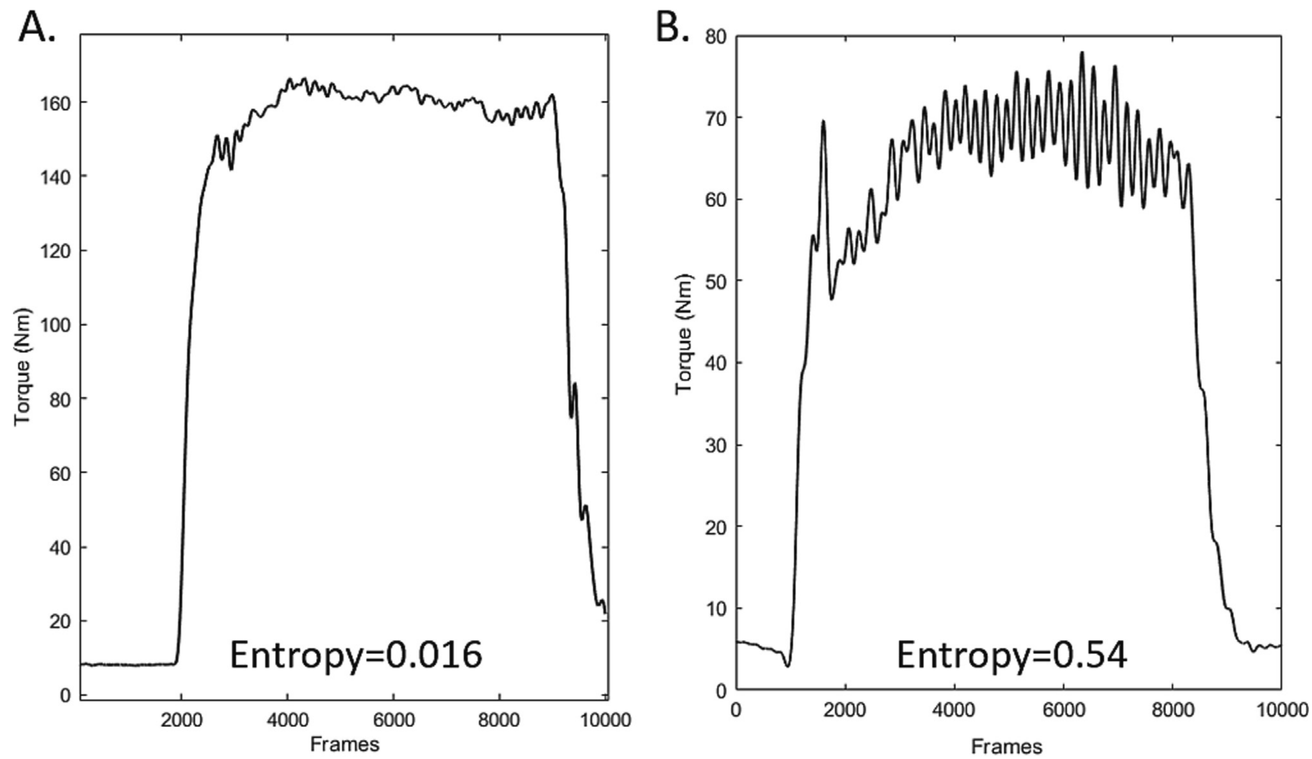

Fig. 1. Representative torque curve and entropy values of the Non-ACL limb (A) and the ACL limb (B) during the maximal voluntary isometric contraction from a single subject. A lower entropy value (A) is represented by smaller fluctuations with more predictable plateaus of the torque signal while a higher entropy value (B) is represented by larger fluctuations in the plateau of the maximal contraction torque signal.

data points (m) that fall within r.[22] A time series with similar distances between data points indicated predictability and results in lower sample entropy values, while large differences between data points indicated complexity, resulting in greater entropy values. A higher entropy value was represented by larger fluctuations in the plateau of the maximal contraction torque signal (less consistent muscle function, more complexity, Fig. 1B), while a lower entropy value is represented by smaller fluctuations with more predictable plateaus (e.g., a sustained muscle contraction) of the torque signal (Fig. 1A).

A post-hoc power analysis utilizing previous literature assessing torque variability after ACL reconstruction was performed using G*Power 3.1, with an indicated effect size of f = 0.35, and it was estimated that 20 individuals were needed to achieve a minimum of 80 % power with $\alpha = 0.05$. Two separate 3 × 2 (time × limb) repeated measures analysis of variance were used to examine changes over time and between limbs for quadriceps SampEn and peak quadriceps torque in statistical software SPSS (version 27, IBM, Armonk, NY, USA). Univariate F tests and Bonferroni multiple comparison procedures were used post hoc in the case of significant interactions. Cohen's d effect sizes (ES) and confidence intervals (CI) were also reported for increased understanding of the data. Calculations of percent change for SampEn and peak torque from baseline (100 − ((baseline data / post-surgery data) ∗ 100)), of each limb separately, were conducted to help provide descriptive analysis of the data. Additionally, Pearson Correlation Coefficients (r) were conducted to better understand the relationship between peak MVIC and quadriceps sample entropy. All data were assessed for normality via Shapiro-Wilks test and found to be normally distributed. Significance was set a priori at $p < 0.05$.

3. Results

All 34 subjects completed testing at every time point. Demographic data for the subjects is presented in Table 1. A significant limb × time interaction was found for quadriceps torque sample entropy ($p < 0.001$; Fig. 2A) and peak MVIC ($p < 0.001$; Fig. 2B). Quadriceps torque sample entropy was higher in the involved limb compared to the uninvolved limb at MID (ACL mean = 0.19 ± 0.14; NonACL mean = 0.08 ± 0.04; ES: 0.97, CI: 0.44, 1.49; $p < 0.001$) and RTA (ACL mean = 0.17 ± 0.12; NonACL mean = 0.07 ± 0.04; ES: 0.91, CI: 0.39, 1.44; $p < 0.001$), indicating greater complexity. Additionally, the involved limb demonstrated an increase in torque sample entropy following surgery from PRE to MID (PRE mean = 0.11 ± 0.08; ES: −0.65, CI: −1.16, −0.14; $p = 0.023$), but not PRE to RTA (ES: −0.51, CI: −1.00, 0.01; $p = 0.169$) or MID to RTA (ES: 0.19, CI: −0.31, 0.69; $p = 0.541$). The uninvolved limb showed a decrease in entropy from PRE to RTA (PRE mean = 0.13 ± 0.11; ES: 0.61, CI: 0.10, 1.12; $p = 0.003$), but no differences between PRE and MID (ES: 0.56, CI: 0.05, 1.06; $p = 0.085$), or MID and RTA (ES: 0.12, CI: −0.38, 0.62; $p = 1.000$).

Quadriceps peak MVIC was lower in the involved limb when compared to the uninvolved limb at PRE (ACL mean = 1.78 ± 0.36; NonACL mean = 2.26 ± 0.37; ES: −1.28, CI: −1.83, −0.74; $p < 0.001$) MID (ACL mean = 1.53 ± 0.44; NonACL mean = 2.39 ± 0.46; ES: −1.84, CI: −2.44, −1.25; $p < 0.001$) and RTA (ACL mean = 1.68 ± 0.52; NonACL mean = 2.23 ± 0.44; ES: −1.11, CI: −1.64, −0.57; $p <$

Table 1
Demographic information for patients. Where appropriate data presented as Mean (SD). BPTB = bone patellar tendon bone graft, HAM = hamstring graft, QUAD = quadriceps tendon graft. When reporting meniscal pathologies, one individual's surgery resulted in both a meniscectomy and a meniscal repair.

Biological sex	19 females, 15 males
Age	16.5 (2.7) yrs
	Range: 14–30 yrs
Mass	75.5 (18.6) kg
Height	1.72 (0.10) m
Pre-injury Tegner score	9.1 (0.9)
Meniscal pathologies	No damage = 17
	Meniscectomy = 4
	Meniscus repair = 11
	Debridement = 3
Graft types	25 BPTB, 6 HAM, 3 QUAD

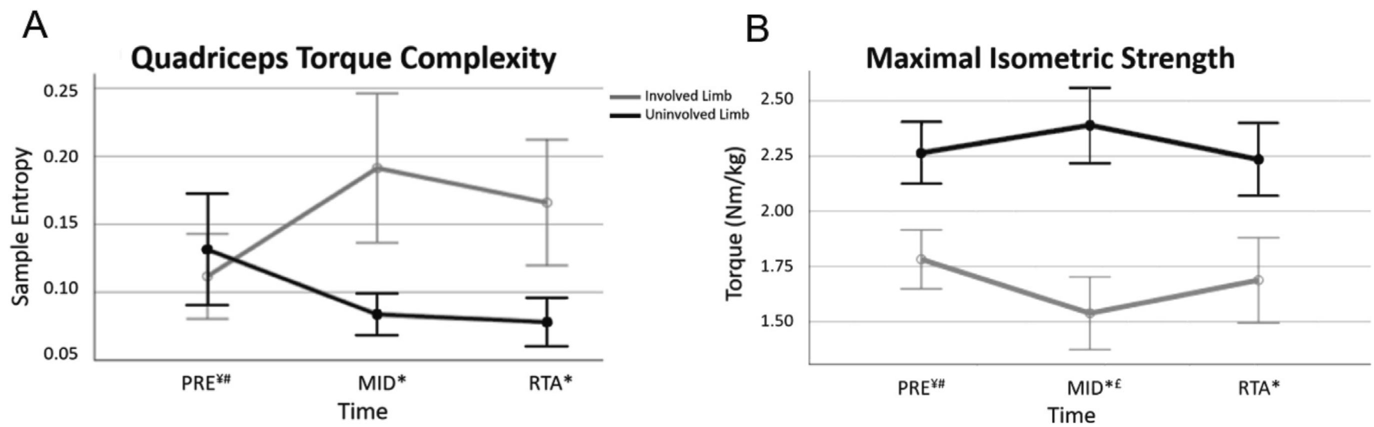

Fig. 2. The gray line representing the involved (ACL) limb, and the black line representing the uninvolved (Non-ACL limb). *Differences between the involved and uninvolved limb, ¥ difference between PRE and MID in the involved limb, # difference between PRE and RTA in the uninvolved limb, £ difference between MID and RTA in the involved limb. A) Quadriceps torque complexity differences from prior to ACLR but after injury (PRE), to approximately halfway through rehab (MID) and at the time participants were cleared to return to physical activity (RTA). B) Maximal isometric strength differences from prior to ACLR but after injury (PRE), to approximately halfway through rehab (MID) and at the time participants were cleared to return to physical activity (RTA).

0.001). Further, peak MVIC of the involved limb was lower at MID compared to PRE (ES: 0.59, CI: 0.08, 1.10; $p = 0.023$) and MID compared to RTA (ES: −0.41, CI: −0.91, 0.10; $p = 0.041$), with no differences between PRE and RTA (ES: 0.21, CI: −0.29, 0.71; $p = 0.865$).

There was no significant relationship between peak MVIC and quadriceps torque sample entropy at any time point for either limb (Table 2).

4. Discussion

The current study is the first to our knowledge to longitudinally assess torque complexity using sample entropy after ACL injury and through ACL reconstruction and rehabilitation. Quadriceps torque complexity was not different between limbs after ACL injury, however, a 42 % increase in torque complexity in the involved limb was detected from pre-surgery to ~5 months after ACL reconstruction, with a 57 % difference between the ACL and non-ACL limbs at this time point. The higher torque complexity in the ACL reconstructed limb persisted at the time individuals returned to physical activity with a 52 % difference noted between limbs. Overall, these findings support that at the time when individuals are cleared to return to physical activity, after ACL reconstruction, they still have altered quadriceps control compared to their uninvolved limb.

The increase in quadriceps torque complexity that we identified in the ACL reconstructed limb compared to the non-injured limb after surgery agrees with prior literature also using non-linear analyses. Bodkin et al.[21] found increased quadriceps isometric torque complexity, assessed using approximate entropy, 6 months after ACL reconstruction when compared to the contralateral limb and compared to healthy individuals. Czaplicki et al.,[19] using wavelet analyses, also noted greater irregularities of isokinetic quadriceps torque six months after ACL reconstruction compared to uninjured controls. Our work also agrees with findings assessed with linear analyses and that showed increased torque variability in individuals after ACL reconstruction compared to healthy individuals, 18 months[14] and 3.5 years[10] after surgery suggesting that even following successful surgery and rehabilitation quadriceps torque complexity/force control is still impaired in the ACL limb. Thus, our work along with others support that quadriceps control is affected by ACL reconstruction and we believe that post-surgical rehabilitation and exercises after formal release from rehabilitation should consider not only strength recovery, but also improving the control/decreasing the complexity of quadriceps contractions.

There are two possible interpretations of our finding of increased torque complexity after ACL reconstruction. One perspective is that greater variability in the torque signal is clinically positive as some have suggested that increased complexity in the torque signal reflects the adaptability of one's motor output in response to the demands of a task.[11,16] In other words, greater complexity may reflect that a person has more neuromuscular strategies being used (e.g. more central and/or peripheral approaches contributing to the excitatory and inhibitory traits of the motor unit pool) to maintain force control and thus are more adaptable to outside perturbations.[11] Research that examines how fatigue affects torque complexity, consistently shows that fatigue decreases torque complexity, and it is often argued that this is a negative consequence because fatigue reduces muscle adaptability and limits the neuromuscular strategies employed to maintain a contraction.[11] On the other hand, it could be postulated that an increase in torque complexity or a less smooth, more variable muscle contraction is the result of impaired motor coordination.[23] Increased variability in the torque signal of ACL patients using other metrics (e.g., coefficient of variation, root mean square error, approximate entropy)[10,13–15,21] is interpreted in this manner and we agree with this interpretation. We would also argue that if increased torque complexity is reflective of improved adaptability then it should not be necessary to employ multiple neuromuscular strategies to complete a relatively simple, uniplanar dynamic activity like an isometric quadriceps contraction.[25] If ACL patients needed to use a variety of approaches to maintain a simple quadriceps contraction, then during a more dynamic and difficult movement (e.g. walking, cutting or landing) we believe that even more strategies would be necessary leading to increased complications.

Table 2
Pearson Correlation Coefficients to understand relationships between peak MVIC and sample entropy at each time, on each limb individually, represented as r (p-value).

	BASE ACL SE	Base NonACL SE	MID ACL SE	MID NonACL SE	RTA ACL SE	RTA NonALC SE
BASE ACL MVIC	0.00 (0.99)					
BASE NonACL MVIC		−0.28 (0.13)				
POST ACL MVIC			−0.20 (0.26)			
POST NonACL MVIC				0.09 (0.61)		
RTA ACL MVIC					−0.29 (0.09)	
RTA NonACL MVIC						−0.32 (0.06)

Anecdotally, the increases in quadriceps torque complexity can be identified clinically as the inability to maintain a consistent level quadriceps torque reading during a maximal voluntary isometric contraction. When visually looking at the torque curve, it is easy to see the increases in complexity by the large fluctuations during the plateau of the MVIC contraction (Fig. 1). We believe this further highlights that the increased complexity noted in the ACL reconstructed limb is not a positive or beneficial phenomenon, but rather represents a lack of quadriceps force control (or an inability to consistently maintain a higher torque). This also may have implications for athletes that are consistently required to perform at high levels, as the inability to sustain a consistent quadriceps contraction could lead to decreased performance and/or increased injury risk. Previous research has identified that after ACL injury those with irregular isokinetic force control are more likely to have poor performance in a single limb hop.[12] If the relationship between torque complexity and performance persists following surgery it may have implications for injury risk, especially for those who have recently been cleared to return to activity, where graft injury or contralateral knee injuries have been shown to be present in almost 30 % of individuals after ACL reconstruction.[26] Future research should continue to explore the relationship between quadriceps force control and performance after ACL reconstruction, as well as interventions capable of improving torque complexity.

The mechanisms resulting in altered torque complexity are unclear, but could be the result of neurophysiological changes occurring to the quadriceps after surgery. Previous research has found alterations in muscle fiber typing after ACL reconstruction.[27,28] Despite serious rehabilitation and targeted strengthening, decreases in type IIA muscle fibers (muscle fibers responsible for maximal strength effort) and increases in type IIA/X fiber types (highly fatigueable muscle fibers) have been reported after ACL reconstruction, at the time patients returned to activity.[28] Previous research has indicated that shift from type IIA to type IIA/X hybrid muscle fiber types is representative of muscle disuse and detraining.[27,28] This change in fiber type may be one reason that the quadriceps muscle struggles to sustain a consistent contraction. Another possible mechanism leading to altered torque complexity after ACL reconstruction may be impaired timing of action potentials. In healthy muscle, the change in force fluctuations has been attributed to timing of action potentials.[23] Davi et al. found slower fascicle shortening velocity in ACL reconstructed limbs compared to uninvolved limbs,[29] thus increased force fluctuations may be a cause of slower action potential firing and shortening velocity that is required to sustain consistent maximal force in the quadriceps. Future studies need to be conducted to better understand the physiological mechanisms and consequences of quadriceps torque complexity after ACL reconstruction.

While greater quadriceps torque complexity was noted after ACL reconstruction, we failed to identify differences between limbs after the ACL injury, prior to the ACL reconstruction. This finding agrees with work by Skurvydas et al.[30] who found no difference in quadriceps isometric torque control (using a nonlinear approach) in ACL-deficient limbs compared to the healthy contralateral limb, but disagrees with prior work by Pua et al.[12] showing decreased isokinetic force control in the ACL-deficient limb compared to the uninjured limb. Pua et al.[12] studied isokinetic contractions where we examined isometric contractions which may help explain differences. Our work does also partially disagree with work by Hollman et al.[31] who used nonlinear fractal scaling to examine sub-maximal isometric quadriceps force control between limbs in ACL-deficient participants. They noted that force steadiness/control was less in the ACL-limb compared to the control limb at 25 % and 35 % of the MVIC, but didn't show differences between limbs at 10 % and 50 % of the MVIC (they also failed to detect between limb differences with coefficient of variation). The lack of between limb differences at 100 % of MVIC in the current study and the 50 % in the Hollman study[31] may suggest that torque or force complexity may not be affected at higher levels of contraction after ACL injury.

While our work and that of some others does suggest that there may not be differences in torque complexity/control in ACL-deficient subjects it is quite possible that there are bilateral changes in torque complexity that may arise after ACL injury that would not be detectable by making between limb comparisons. Recent work by Qiu et al.[32] supports this idea as they found differences between ACL limb and healthy control subjects in various measurements of quadriceps function (i.e. strength, activation, and rate of torque development). More work needs to be conducted to determine if ACL injury alters quadriceps torque complexity bilaterally using nonlinear analytic techniques. Furthermore, it would be of value to assess the sensitivity of the different metrics utilized to quantify torque complexity in ACL patients given that a variety of measures are used[12,21,31] and there is no agreement on which is best. We would argue that the literature preliminarily supports sample entropy as a superior approach to some of the others (e.g., approximate entropy) as it a more accurate for shorter data series, like those seen with quadriceps isometric contractions and has also been shown to reduce biases.[22]

A decrease in the non-ACL limb torque complexity was noted for our participants at the time of return to physical activity when compared to after ACL injury. This significant decrease in quadriceps torque complexity in the non-ACL limb may be a result of successful rehabilitation. Alternatively, it is possible that the decrease in non-ACL limb torque complexity is the result of the non-ACL limb adapting or compensating to make up for impairments in the ACL limb.

While not surprising, quadriceps strength was lower in the ACL-reconstructed limb compared to the uninjured limb at all-time points. Further, strength in the ACL-reconstructed limb decreased from presurgery to 5 months after reconstruction, but increased from 5 months after surgery to the time when patients returned activity. The results of lower strength in the affected limb after ACL injury and reconstruction are supported by a plethora of studies.[4] While quadriceps strength of the ACL-reconstructed limb does appear to improve throughout the course of rehabilitation, it does not have equivalent strength to that of the uninjured limb at the time of return to activity. Quadriceps strength deficits in conjunction with increased torque complexity suggest that force production and control are negatively influenced by ACL injury and reconstruction and should continue to be a focus for clinicians.

The lack of correlation between quadriceps peak torque and quadriceps torque complexity indicates that examining torque complexity provides additional information related to quadriceps function that is not available from traditional discrete, magnitude-based (e.g. peak torque) analyses. The poor relationship between these variables also suggests that it is possible for a patient to present with a strong quadriceps, but still have poor control of the muscle.

There are limitations to this study that are important to recognize. First, our data is limited to the quadriceps torque signal to assess the quality of the muscle contraction. Utilizing electromyography during maximal torque production would allow us to glean important information on how muscle electrical activity is changing throughout rehabilitation, especially on an individual muscle contribution level. Additionally, this analysis did not include a healthy uninjured control group for comparison. Future work including a control group would better help us understand how torque complexity may change in both the uninvolved and involved limbs immediately after injury, and over time. It is also important to note that while all participants were skeletally mature and none underwent a pediatric ACL reconstruction, we do include a wide age range (14–30) in our study that did not focus on only an adult or adolescent population. Similarly, we include all, bone patellar tendon bone, quadriceps tendon and hamstring graft types in this study, but with a higher number of patellar tendon grafts ($n = 25$; Table 1) which may influence our results.

5. Conclusion

In conclusion, we identified that following ACL reconstruction there are increases in quadriceps torque complexity and peak torque between limbs that continue at least until the time when participants are cleared to return to activity. These increases in quadriceps torque complexity are likely representative of patients' inability to sustain a consistent, controlled quadriceps contraction.

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Confirmation of Ethical Compliance

This study met and followed all ethical guidelines via approval by the University of Michigan Institutional Review Board under HUM#00110455.

CRediT authorship contribution statement

Alexa K. Johnson: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Kazandra M. Rodriguez:** Conceptualization, Software, Resources, Validation, Investigation, Data curation, Writing – review & editing, Visualization. **Adam S. Lepley:** Validation, Formal analysis, Writing – review & editing, Visualization. **Riann M. Palmieri-Smith:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Interest Statement

Alexa K. Johnson, Kazandra Rodriguez, Adam Lepley, and Riann Palmieri-Smith all report no competing interests, conflicts of interest, or other kinds of associations.

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